A cellular automata model for predicting fire spread

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Abstract

A new model for predicting fire propagation in homogeneous environments, based on the transfer of a fractional burned area, is introduced in this paper. A particular type of discrete dynamical system, called a hexagonal bidimensional cellular automaton, is used in order to design the model. The simulations obtained are found to be in agreement with the experience of fire spreading in real forests.
Keywords: fire spreading, hexagonal cellular automata, mathematical modelling.

1 Introduction

As is well-known, forest fires is one of the major ecological agents in forest since they alter the structure and composition of forest and adversely affect human health and the supply of goods and services on which communities depend. Consequently, the study of forest fires have received increased public attention by the scientific community. Specifically, several mathematical models have been appeared in the literature in order to study the spreading of forest fire fronts. The efforts to model the growth of the fire front by means of mathematical models can be classified according to two approaches: vector models and cellular automata models. Vector models assume that the fire front spreads according to a well-defined growth law, and take a standard geometrical shape (see, for example, Vakalis *et al.* [6]). On the other hand, several types of cellular automata models for fire spreading have been introduced in the literature (see Karafyllidis [1, 2]).

The main goal of this work is to introduce a new discrete mathematical model for predicting the behaviour of the forest fire front based on a two-dimensional hexagonal cellular automata. Basically, it consists of a two-dimensional hexagonal array of a finite number of identical objects called cells, interacting locally with each other. Each cell can assume a state from a finite or infinite state set, which changes at every time step according to a fixed rule. The variables of such function are the states of some neighbour cells at the previous time step. The proposed model divides the forest into regular hexagonal areas, each one of them stands for a cell. The state of each cell is a real number from the interval [0,1]. If it is 0, then the cell is unburned, and if the state is 1, the cell is totally burned out. The evolution rule linearly depends on thirteen variables: the states of the twelve nearest cells to a given one and its own state. These variables are affected by some parameters which stand for wind speed and the topography.

The rest of the paper is organized as follows: In Section 2 a review of bidimensional cellular automata is given; the proposed model is introduced in Section 3; some graphical simulations are shown in Section 4; and, finally, the conclusions are presented in Section 5.

2 A review of cellular automata

In a precise way, a cellular automaton, CA for short, (see, for example, Sarkar [4], and Toffoli *et al.* [5]) can be defined as a 4-uplet $A= (C, S, V, f)$, where C is the cellular space formed by a bidimensional array of $r\times c$ cells, $\{(a,b), 1 \le a \le r, 1 \le b \le c\}$, such that each of which can assume a state. We can consider two types of cellular spaces: square cellular spaces, formed by a bidimensional array of identical square areas, each one of them stands for a cell, and hexagonal cellular spaces, where each cell stands for a regular hexagonal area. This work deals with this second type.

The state of each cell is an element of a state set, S. Furthermore, the state of a particular cell (a,b) at time t is denoted by $s_{ab}^{(t)}$.

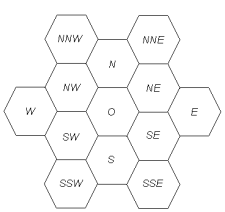

Figure 1: The neighborhood of the cell O.

The set of indices of the CA is the ordered finite subset $V \subset \mathbf{Z}\times\mathbf{Z}$, $|V| = m$, such that for every cell (a,b), its neighborhood $V_{(a,b)}$ is the ordered set of m cells given by $V_{(a,b)}= \{(a+\alpha_1, b+\beta_1),\ldots, (a+\alpha_m,b+\beta_m) : (\alpha_i, \beta_i)\in V\}$.

Depending on the process to be modelled, one can choose an appropriate neighbourhood. In this work, the neighbourhood of a particular cell $O = (a,b)$ is given by the set

$$V_O = \{O, N, NE, SE, S, SW, NW, NNE, E, SSE, SSW, W, NNW\}, \quad (1)$$

which is shown in Figure 1.

Due to the main features of the model for predicting fire spread proposed below, we must distinguish two types of neighbour cells of the cell O: near neighbour cells, $V^n = \{N, NE, SE, S, SW, NW\}$; and distant neighbour cells, $V^d = \{NNE, E, SSE, SSW, W, NNW\}$. Moreover, each distant neighbour cell, $(\alpha, \beta) \in V^d$, has two associated near neighbour cells: $(a+\alpha^+, b+\beta^+)$ and $(a+\alpha^-, b+\beta)$, that are given by those with common sides. For example, the associated near neighbor cells of NNE are N and NE, and so on. Furthermore, in this case the set defining the neighborhood, $V \subset \mathbf{Z}\times\mathbf{Z}$, depends on the cell $O = (a,b)$ to be considered. That is, if b is odd, then (see Figure 2), $V = \{(0,0), (-1,0), (0,1), (1,1), (1,0), (1,-1), (0,-1), (-1,1), (0,2), (2,1), (2,1),(0,-2), (-1,-1)\}$;

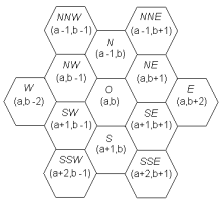

Figure 2: The labelled neighborhood when b is odd.

On the other hand, if b is even, then (see Figure 3), $V = \{(0,0), (-1,0), (-1,1), (0,1), (1,0), (0,-1), (-1,-1), (-2,1), (0,2), (1,1), (1,-1), (0,-2), (-2,-1)\}$.

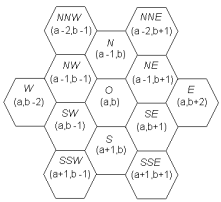

Figure 3: The labelled neighborhood when b is even.

For the sake of simplicity and, since it does not affect to the computations below, we will consider the generic neighborhood V_O given in eqn. (1).

The CA evolves deterministically in discrete time steps, changing the states of all cells according to a local transition function: f: $S^{13} \rightarrow S$. The updated state of the cell (a,b) depends on the thirteen variables of the local transition function,

which are the previous states of the cells constituting its neighborhood, that is, $s_{ab}^{(t+1)} = f\left(s_{(a+\alpha_1,b+\beta_1)}^{(t)},\ldots,s_{(a+\alpha_{13},b+\beta_{13})}^{(t)}\right)$, with $(\alpha_i,\beta_i)\in V$, for every $1\le i\le 13$. Moreover, the matrix, $C^{(t)} = (s_{ij}^{(t)})$, is called the configuration at time t of the CA, with $C^{(0)}$ being the initial configuration. Since the number of cells of the CA is finite, boundary conditions must be considered in order to assure the well-defined dynamics of such CA. In this work null boundary conditions are considered: If $(a,b)\notin \{(u,v),\ 1\le u\le r, 1\le v\le c\}$, then $s_{ab}^{(t)} = 0$.

Linear CA are those whose local transition functions are linear with real coefficients. Note that, in this case, the state set is infinite: $S = \mathbf{R}$. Nevertheless, depending on the process to be modeled, the state set of the CA is usually finite and, consequently, a discretization function, g:$\mathbf{R}\rightarrow S$, must be considered. Thus, in this case, the local transition function is as follows:

$$s_{ab}^{(t+1)} = g\left(\sum_{(\alpha,\beta)\in V}\mu_{\alpha\beta}s_{a+\alpha,b+\beta}^{(t)}\right),\quad \mu_{\alpha\beta}\in\mathbf{R}.$$

3 The CA-based model for fire spreading

3.1 The state set

A forest area can be interpreted as the hexagonal cellular space of a CA by simply dividing it into a bidimensional array of identical regular hexagonal areas of side length L. Obviously, each one of this areas stands for a cell of the CA. The state of a cell (a,b) at a time t, is defined as follows:

$$s_{ab}^{(t)} = \frac{\text{burned out area of } (a,b) \text{ at } t}{\text{total area of } (a,b)} = \frac{\text{burned out area of } (a,b) \text{ at } t}{3\sqrt{3}L^2/2}\ .$$

If $s_{ab}^{(t)} = 0$, then the cell (a,b) is said to be unburned at time t; if $0 < s_{ab}^{(t)} < 1$, the cell (a,b) is partially burned out at time t, and finally, if $s_{ab}^{(t)} = 1$, the cell is said to be completely burned out at time t. The evolution of the CA used in the proposed model supposes that the state of the cell (a,b) at time t+1 linearly depends on the states of its neighbor cells at time t.

In the proposed model, we will consider a finite state set of order \$k\$, that is, $S = \{q_1,\ldots,q_k\}$, with $q_i\in\mathbf{R}$ for every i and $0\le q_1\le\ldots\le q_k\le 1$. Then, a suitable discretization function, g:[0,1]$\rightarrow S$, must be incorporated to the model as follows:

$$s_{ab}^{(t+1)} = g\left(s_{ab}^{(t)} + \sum_{(\alpha,\beta)\in V^n}\mu_{\alpha\beta}^{(a,b)}s_{a+\alpha,b+\beta}^{(t)} + \sum_{(\alpha,\beta)\in V^d}\mu_{\alpha\beta}^{(a,b)}s_{a+\alpha,b+\beta}^{(t)}\right), \tag{2}$$

where $\mu_{\alpha\beta}^{(a,b)}\in\mathbf{R}$ are parameters involving some physical magnitudes of the cells. Note that from eqn. (2), it is possible that the value $s_{ab}^{(t)}$ to be greater than 1. In this case, $s_{ab}^{(t)}$ is taken to be equal to 1.

3.2 Incorporating weather and topography conditions

Each cell O can be endowed with three parameters: The rate of fire spread, R, which is the same for all cells as the forest is homogeneous, the wind speed, W_O, and the height H_O. Specifically, the effect of the wind on the cell O is given by the set $W_O = \{\omega^O_{\alpha\beta}, (\alpha,\beta) \in V\}$, where $\omega^O_{\alpha\beta} > 0$, in such a way that if no wind is blowing on O, then $\omega^O_{\alpha\beta} = 1$ for every $(\alpha,\beta) \in V$; if the wind is blowing from North to South, then the coefficients ω^O_{NW}, ω^O_{NNW}, ω^O_{N}, ω^O_{NNE} and ω^O_{NE} must be larger than the rest of coefficients, and so on. The value of such coefficients stands for the magnitude of the wind. Furthermore, the height differences between various points in a forest also affect the fire spreading. As is well known, the fires show a higher rate of spread when they climb an upward slope, whereas fires show a smaller rate of spread when they descend a downward slope. The height influence of a near neighbor cell, $(a+\alpha, b+\beta)$, with $(\alpha,\beta) \in V$, on a cell $O = (a,b)$, is given by $h^O_{\alpha\beta} = \phi\left(H_{(a,b)} - H_{(a+\alpha,b+\beta)}\right)$. The function $y = \phi(x)$ must be determined according to the characteristic of the forest, and, also, it must satisfy the following conditions: If $x<0$, then $0< \phi(x)<1$; if $x = 0$, $\phi(x) = 1$, and, finally, if $x>0$, $\phi(x)>1$. Note that the first condition establishes that if $H_{(a,b)} > H_{(a+\alpha,b+\beta)}$, the fire increases its rate of spread; the second condition states that when $H_{(a,b)} = H_{(a+\alpha,b+\beta)}$, the topography does not affect the fire spread; and the third condition establishes that if $H_{(a,b)} < H_{(a+\alpha,b+\beta)}$, the fire restrains its spreading.

On the other hand, the height influence of a distant neighbor cell is affected by the influence of its associated near neighbor cells. For example, if the distant neighbor cell is *NNE*, then:

$$h^O_{NNE} = \frac{1}{4}\left(\phi\left(H_O - H_N\right) + \phi\left(H_N - H_{NNE}\right) + \phi\left(H_O - H_{NE}\right) - \phi\left(H_{NE} - H_{NNE}\right)\right).$$

Note that if the forest is flat (all cells have the same height), then $\phi(x) = 1$ and consequently $h^{(a,b)}_{\alpha\beta} = 1$ for every cell (a,b) and $(\alpha,\beta) \in V$. Consequently, $\mu^{(a,b)}_{\alpha\beta} = \omega^{(a,b)}_{\alpha\beta} \cdot h^{(a,b)}_{\alpha\beta}$, where $\omega^{(a,b)}_{\alpha\beta}$ and $h^{(a,b)}_{\alpha\beta}$ stand for the wind influence and the height influence respectively of the neighbor cell $(a+\alpha,b+\beta)$ on (a,b).

3.3 Determining the time step

In the proposed model, the size of the time step, $\tilde{t}$, is considered to be equal to the time needed for a one and only near neighbor cell to burn the main cell out. Consequently, the rate of fire spread of the cells, R, determines $\tilde{t}$, and it is assumed to be given by a different model (see, for example, [3]). As a consequence, it is easy to check that if all cells in the neighborhood of O are unburned at time t except for only one near neighbor cell, say for example N,

then the time needed for O to be completely burned out is $\tilde{t} = \sqrt{3}L/R$ (see Figure 4).

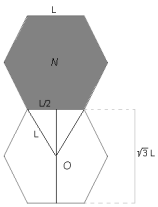

Figure 4: The size of the discrete step of time.

Thus, if all cells in the neighborhood of O are unburned at time t except only one adjacent cell, which is completely burned out, then at time t+1, the cell O is completely burned out: $s_O^{(t+1)} = 1$. Furthermore, if the only completely burned out neighbor cell at time t, is a distant neighbor cell of O, say *NNE*, then $s_O^{(t+1)} = \lambda < 1$. That value is calculated as follows. In a step time $\tilde{t}$, the near neighbor cells, N and *NE*, and a little portion (a circular sector) of O will be burned out. Specifically, as the distance covered by the fire front in $\tilde{t}$ with speed R is $\sqrt{3}L$, then the radius of the circular sector of O burned out is $\sqrt{3}L - L$ (see Figure 5).

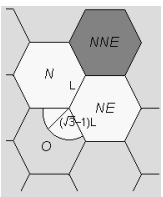

Figure 5: The calculus of λ.

As a consequence, the burned out area of the cell O is

$$\frac{\pi\left(\sqrt{3}-1\right)^2 L^2 \frac{\pi}{3}}{2\pi} = \frac{2-\sqrt{3}}{3}\pi L^2 .$$

So, if all neighbor cells of O are unburned at time t, except a distant neighbor cell which is fully burned out, then

$$s_O^{(t+1)} = \lambda = \frac{\frac{2-\sqrt{3}}{3}\pi L^2}{\frac{3\sqrt{3}}{2}L^2} = \frac{4\sqrt{3}-6}{27}\pi \approx 0.1080.$$

3.4 The local transition function

Taking into account the results from the previous sections, one can conclude that the local transition function of the hexagonal cellular automata for predicting forest fire spreading is as follows:

$$s_{ab}^{(t+1)} = g\left(s_{ab}^{(t)} + \sum_{(\alpha,\beta)\in V^n} \mu_{\alpha\beta}^{(a,b)} s_{a+\alpha,b+\beta}^{(t)} + \sum_{(\alpha,\beta)\in V^d} \mu_{\alpha\beta}^{(a,b)} s_{a+\alpha,b+\beta}^{(t)} \right),$$

with $\mu_{\alpha\beta}^{(a,b)} = \omega_{\alpha\beta}^{(a,b)} \cdot h_{\alpha\beta}^{(a,b)}$, where

$$h_{\alpha\beta}^{(a,b)} = \begin{cases} \phi\left(H_{(a,b)} - H_{(a+\alpha,b+\beta)}\right), \text{ if } (\alpha,\beta)\in V^n \\ \frac{1}{4}\left[\phi\left(H_{(a,b)} - H_{(a+\alpha^+,b+\beta^+)}\right) + \phi\left(H_{(a+\alpha^+,b+\beta^+)} - H_{(a+\alpha,b+\beta)}\right)\right. \\ \left.\phi\left(H_{(a,b)} - H_{(a+\alpha^-,b+\beta^-)}\right) + \phi\left(H_{(a+\alpha^+,b+\beta^+)} - H_{(a+\alpha,b+\beta)}\right)\right], \text{ if } (\alpha,\beta)\in V^d \end{cases}$$

and g:[0,1]→S. Note that if there are no wind and topography conditions, the explicit expression of the transition function is as follows:

$$s_{ab}^{(t+1)} = g\left(s_{ab}^{(t)} + \sum_{(\alpha,\beta)\in V^n} s_{a+\alpha,b+\beta}^{(t)} + \lambda \sum_{(\alpha,\beta)\in V^d} s_{a+\alpha,b+\beta}^{(t)} \right).$$

4 Testing the proposed model

A good model for predicting forest fire spreading must pass some tests. In this work, we will consider four basic tests: (1) Modelization of a fire front spreading on a flat forest without wind conditions; (2) Modelization of a fire front spreading on a flat forest with wind conditions; (3) Modelization of a fire front spreading on a non-flat forest without wind conditions; and (4) Modelization of a fire front spreading on a non-flat forest with wind conditions. If an homogeneous flat forest without wind conditions is considered, the model must yield a circular fire front. If there are weather conditions, the wind speed and direction must affect the forest fire front. Furthermore, if the forest is non-flat, the topography conditions must be reflected in the dynamics of fire front since, as was mentioned above, fires show a higher rate of spread when they climb up an upward slope, whereas fires show a smaller rate of spread when they descend a downward slope.

An algorithm using the computer algebra system Mathematica has been implemented for the computational and graphical simulations of the fire fronts. The hypothetical forests used are modeled by means of a bidimensional array of 100×100 cells. In the initial configuration, there is only one cell (the central cell) burned out, whereas the rest of the cells are unburned, and 20 evolutions are calculated. In the figures below, only the configurations at times t = 0, 5, 10, 15, 20 are shown. Furthermore, in the simulations, the state set used is

$S = \{q_1 = 0, q_2 = 0.1, \ldots, q_{11} = 1\}$, and consequently, the discretization function, g:[0,1]→S, will be

$$g(t) = \begin{cases} 0, \text{if } t = 0 \\ q_i, \text{if } \frac{i-1}{10} \le t \le \frac{i}{10} \\ 1, \text{if } t = 1 \end{cases}$$

Moreover, the states of the cells are represented according to the following color code (see Figure 6).

Figure 6: The color code.

Firstly, suppose that the forest is flat, and there are no wind conditions. Then, as is shown in Figure 7, the fire front is circular.

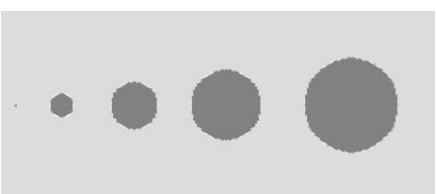

Figure 7: Simulation in a flat forest without wind conditions.

If the forest is flat and the wind is blowing from north to south according to the wind front given in Figure 8), then the simulation obtained is shown in Figure 9.

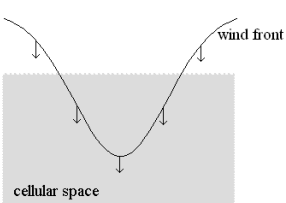

Figure 8: The wind front.

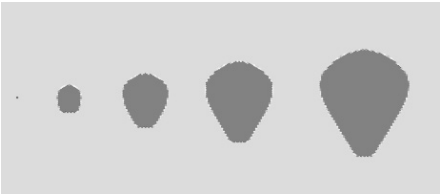

Figure 9: Simulation in a flat forest with wind conditions.

Now, suppose that the forest is non-flat, say for example, $H_{(a,b)} = b$ for each cell (a,b), and take, for example, the function

$$\phi(x) = \begin{cases} e^x, \text{if } x \le 0 \\ 1 + \dfrac{\sqrt{x}}{10}, \text{if } x > 0 \end{cases}$$

If there are not wind conditions, the spreading of the fire forest is shown in Figure 10, and if the wind blows from north to south according to the above non-linear front, the fire forest evolves as is shown in Figure 11.

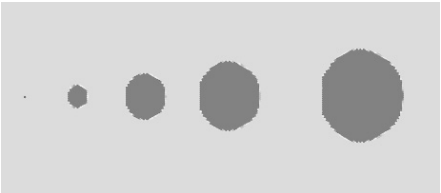

Figure 10: Simulation in a non-flat forest without wind conditions.

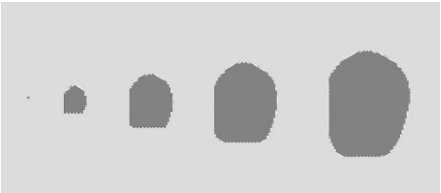

Figure 11: Simulation in a non-flat forest with wind conditions.

5 Conclusions

In this work, a new model for the prediction of forest fire spreading has been introduced. It is based on hexagonal cellular automata and incorporates weather

and land topography conditions. The algorithm seems to be very efficient and it is easily implemented in any computer algebra system. The model is applied to four basic cases depending on the weather and topography conditions. All the graphical models obtained are found to be in agreement with the experience of fire spreading in real forests.

Acknowledgement

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